

**САНКТ-ПЕТЕРБУРГСКИЙ НАЦИОНАЛЬНЫЙ
ИССЛЕДОВАТЕЛЬСКИЙ УНИВЕРСИТЕТ ИТМО**

Дисциплина: Бэк-энд разработка

Отчет

Лабораторная работа №4
«Развёртывание приложения на удалённом сервере»

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2026 г.

Задача

Настроить удалённый сервер, развернуть на нём приложение и настроить nginx как обратный прокси.

Вариант 5. Сервис для обмена рецептами и кулинарных блогов.

Ход работы

1. Сервер

Арендован VPS с Ubuntu 22.04, 888 МБ памяти и диском 39 ГБ. Docker на сервере уже был. На нём работают контейнеры VPN (amnezia) и контейнер nginx-nginx-1, их не трогали. Файла подкачки не было.

```
[dimaapleev@MacBook-Air-Dima-2 ~ % ssh root@140.150.226.124
root@140.150.226.124's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.0-191-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.
Last login: Mon Sep 21 19:22:18 2026 from 157.228.152.149
root@localhost:~# cat /etc/os-release | head -2
PRETTY_NAME="Ubuntu 22.04.5 LTS"
NAME="Ubuntu"
root@localhost:~# free -h
              total        used        free      shared  buff/cache   available
Mem:           888Mi       336Mi       142Mi         4.0Mi        409Mi       400Mi
Swap:              0B           0B           0B
root@localhost:~# df -h /
Filesystem      Size  Used Avail Use% Mounted on
/dev/mapper/hk-root  39G   7.3G   30G   20% /
root@localhost:~# docker ps --format '{{.Names}}  {{.Ports}}'
amnezia-awg2  0.0.0.0:35609->35609/udp, [::]:35609->35609/udp
amnezia-openvpn  0.0.0.0:41065->41065/tcp, [::]:41065->41065/tcp
amnezia-wireguard  0.0.0.0:40581->40581/udp, [::]:40581->40581/udp
nginx-nginx-1
```

Рисунок 1. Сервер до развёртывания

2. Как устроен nginx на сервере

nginx уже работает в контейнере из образа jonasal/nginx-certbot и сам получает сертификат HTTPS. Он слушает порты 80 и 443 и запущен в сети хоста (host), поэтому видит порты самого сервера. Конфиг сайта лежит на сервере в папке /data/nginx/user_conf.d и подключён в контейнер.

```
dimaapleev — root@localhost: /opt/recipes — zsh — 80x36

amnezia-openvpn 0.0.0.0:41065->41065/tcp, [::]:41065->41065/tcp
amnezia-wireguard 0.0.0.0:40581->40581/udp, [::]:40581->40581/udp
nginx-nginx-1
[root@localhost:~# sudo ss -tulpn | grep -E ':(80|443|3004|15672)\s'
tcp LISTEN 0 511 0.0.0.0:443 0.0.0.0:* users:((
"nginx",pid=219919,fd=6),("nginx",pid=1209,fd=6))
tcp LISTEN 0 511 0.0.0.0:80 0.0.0.0:* users:((
"nginx",pid=219919,fd=8),("nginx",pid=1209,fd=8))
tcp LISTEN 0 511 [::]:443 [::]:* users:((
"nginx",pid=219919,fd=7),("nginx",pid=1209,fd=7))
tcp LISTEN 0 511 [::]:80 [::]:* users:((
"nginx",pid=219919,fd=9),("nginx",pid=1209,fd=9))
[root@localhost:~# docker inspect nginx-nginx-1 --format '{{.HostConfig.NetworkMo
de}}'
host
[root@localhost:~# docker inspect nginx-nginx-1 --format '{{range .Mounts}}{{.Sou
rce}} -> {{.Destination}}{{"\n"}}{{end}}'
/var/lib/docker/volumes/nginx_secrets/_data -> /etc/letsencrypt
/data/nginx/user_conf.d -> /etc/nginx/user_conf.d
/data/nginx/letsencrypt -> /var/www/letsencrypt

[root@localhost:~# uname -m
x86_64
[root@localhost:~# ls -la /data/nginx/user_conf.d
total 12
drw-r----- 2 root root 4096 Sep  7 16:02 .
drw-r----- 4 root root 4096 Sep  7 16:02 ..
-rw-r--r-- 1 root root 1362 Sep  7 16:02 django37328.hostkey.in.conf
[root@localhost:~# cat /data/nginx/user_conf.d/*.conf
map $http_upgrade $connection_upgrade {
    default upgrade;
    ''      close;
}

server {
    listen 443 ssl default_server reuseport;
```

Рисунок 2. Порты nginx, режим сети и папки конфигов

В конфиге все запросы к домену передавались демо приложению хостинга на порту 8000 (строка `proxy_pass`). Её мы и заменили на наш BFF.

```
dimaapleev — root@localhost: /opt/recipes — -zsh — 80x36

# Domain names this server should respond to.
server_name django37328.hostkey.in;

# Load the certificate files.
ssl_certificate      /etc/letsencrypt/live/django37328.hostkey.in/fullcha
in.pem;
ssl_certificate_key  /etc/letsencrypt/live/django37328.hostkey.in/privkey
.pem;
ssl_trusted_certificate /etc/letsencrypt/live/django37328.hostkey.in/chain.p
em;

# Load the Diffie-Hellman parameter.
ssl_dhparam /etc/letsencrypt/dhparams/dhparam.pem;

location ^~ /.well-known/acme-challenge/ {
    root /var/www/letsencrypt;
    default_type "text/plain";
}

location / {
    proxy_set_header X-Forwarded-Host $host;
    proxy_set_header X-Forwarded-Server $host;
    proxy_set_header X-Real-IP $remote_addr;
    proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
    proxy_pass http://django37328.hostkey.in:8000;
    real_ip_header X-Forwarded-For;

    # Websockets
    proxy_http_version 1.1;
    proxy_set_header Upgrade $http_upgrade;
    proxy_set_header Connection $connection_upgrade;
    proxy_set_header X-Scheme $scheme;

    proxy_buffering off;
}

[root@localhost:~#docker inspect nginx-nginx-1 --format '{{.Config.Image}}' {{in]
```

Рисунок 3. Конфиг сайта в nginx до изменения

3. Подготовка

Памяти мало для сборки четырёх образов, поэтому добавлен файл подкачки на 2 ГБ и включён при загрузке сервера. Для копирования проекта установлен rsync. Swap виден на рисунке 7.

4. Копирование проекта

Папка lab2 скопирована с компьютера в /opt/recipes. Без node_modules, локальных .env, override файла и отчётов.

```
rsync -av --exclude node_modules --exclude .env \
--exclude docker-compose.override.yml --exclude '*.docx' \
./ root@140.150.226.124:/opt/recipes/
```

5. Секреты

На сервере создан свой .env с новыми случайными ключами (openssl rand). Доступ к файлу только у root (chmod 600). На рисунке видны только имена ключей, без значений.

```
dimaapleev — root@localhost: /opt/recipes — -zsh — 80x36

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
[root@localhost:~# ls /opt/recipes ]
README.md  docker-compose.yml  package.json  social-service
bff-service package-lock.json  recipe-service user-service
[root@localhost:~# cd /opt/recipes ]
root@localhost:/opt/recipes# cat > .env <<EOF
# секреты для docker-compose на сервере
DB_PASSWORD=$(openssl rand -hex 16)
JWT_SECRET_KEY=$(openssl rand -hex 32)
INTERNAL_API_KEY=$(openssl rand -hex 24)
EOF
[root@localhost:/opt/recipes# chmod 600 .env ]
[root@localhost:/opt/recipes# cut -d= -f1 .env ]
# секреты для docker-compose на сервере
DB_PASSWORD
JWT_SECRET_KEY
INTERNAL_API_KEY
[root@localhost:/opt/recipes# ls /opt/recipes ]
README.md  docker-compose.yml  package.json  social-service
bff-service package-lock.json  recipe-service user-service
[root@localhost:/opt/recipes# docker compose version ]
Docker Compose version v5.5.1
[root@localhost:/opt/recipes# docker compose -f docker-compose.yml build user-service]
vice
[+] Building 375.8s (14/14) FINISHED
=> [internal] load local bake definitions 0.0s
=> => reading from stdin 526B 0.0s
=> [internal] load build definition from Dockerfile 0.2s
=> => transferring dockerfile: 203B 0.0s
=> [internal] load metadata for docker.io/library/node:20 2.1s
```

Рисунок 4. Секреты и начало сборки

6. Сборка и запуск

Образы собраны по одному, чтобы хватило памяти. Запуск только по основному файлу (`-f docker-compose.yml`), поэтому порты сервисов закрыты и наружу смотрит только BFF на `127.0.0.1:3004`. Compose скачал образы `postgres` и `rabbitmq`, создал сеть и тома, дождался готовности баз и RabbitMQ и запустил сервисы.

```
dimaapleev — root@localhost: /opt/recipes — zsh — 80x36

=> [6/7] COPY src ./src 0.1s
=> [7/7] RUN npm run build 15.2s
=> exporting to image 61.2s
=> => exporting layers 46.3s
=> => exporting manifest sha256:64a933b494b109ccfd3ed2f954c2ba0b33f3a255 0.0s
=> => exporting config sha256:e1faa63ca6f7b0f76fd5682b2fa6ee518e50c32063 0.0s
=> => exporting attestation manifest sha256:f88d2ce21b61fd8a8a78bae2532b 0.0s
=> => exporting manifest list sha256:080b1cb5175db371cd3ea86d2dc943a33a9 0.0s
=> => naming to docker.io/library/recipes-bff-service:latest 0.0s
=> => unpacking to docker.io/library/recipes-bff-service:latest 14.4s
=> resolving provenance for metadata file 0.2s
[+] build 1/1
✓ Image recipes-bff-service Built 270.7s
[root@localhost:/opt/recipes# docker images | grep recipes ]
recipes-bff-service:latest 080b1cb5175d 1.9GB 487MB
recipes-recipe-service:latest 8b220c35f754 1.91GB 487MB
recipes-social-service:latest 630aa8f07667 1.91GB 487MB
recipes-user-service:latest 64afb5837262 1.9GB 487MB
[root@localhost:/opt/recipes# cd /opt/recipes ]
[root@localhost:/opt/recipes# docker compose -f docker-compose.yml up -d ]
[+] up 42/42
✓ Image postgres:16 Pulled 121.8s
✓ Image rabbitmq:3-management Pulled 92.3s
✓ Network recipes_default Created 0.5s
✓ Volume recipes_recipe-db-data Created 0.1s
✓ Volume recipes_social-db-data Created 0.1s
✓ Volume recipes_user-db-data Created 0.0s
✓ Container recipes-user-db-1 Healthy 18.8s
✓ Container recipes-social-db-1 Healthy 27.2s
✓ Container recipes-recipe-db-1 Healthy 17.2s
✓ Container recipes-rabbitmq-1 Healthy 41.2s
✓ Container recipes-user-service-1 Started 19.6s
✓ Container recipes-recipe-service-1 Started 42.9s
✓ Container recipes-social-service-1 Started 41.3s
✓ Container recipes-bff-service-1 Started 45.5s
[root@localhost:/opt/recipes# docker compose -f docker-compose.yml ps ]
```

Рисунок 5. Готовые образы и запуск контейнеров

Все восемь контейнеров работают, базы и RabbitMQ в состоянии healthy. В логах видно, что сервисы создали таблицы, а social-service подключился к очереди после нескольких попыток.

```
dimaapleev — root@localhost: /opt/recipes — zsh — 80x36

✓ Container recipes-bff-service-1 Started 45.5s
[root@localhost:/opt/recipes# docker compose -f docker-compose.yml ps]
NAME                IMAGE              STATUS              COMMAND              PORTS              SER
VICE              CREATED              STATUS              COMMAND              PORTS              SER
recipes-bff-service-1 recipes-bff-service "docker-entrypoint.s..." bff
-service          About a minute ago Up 22 seconds        127.0.0.1:3004-
>3004/tcp
recipes-rabbitmq-1 rabbitmq:3-management "docker-entrypoint.s..." rab
bitmq            About a minute ago Up About a minute (healthy) 4369/tcp, 5671-
5672/tcp, 15671/tcp, 15691-15692/tcp, 25672/tcp, 127.0.0.1:15672->15672/tcp
recipes-recipe-db-1 postgres:16        "docker-entrypoint.s..." rec
ipe-db           About a minute ago Up About a minute (healthy) 5432/tcp
recipes-recipe-service-1 recipes-recipe-service "docker-entrypoint.s..." rec
ipe-service      About a minute ago Up 23 seconds
recipes-social-db-1 postgres:16        "docker-entrypoint.s..." soc
ial-db           About a minute ago Up About a minute (healthy) 5432/tcp
recipes-social-service-1 recipes-social-service "docker-entrypoint.s..." soc
ial-service      About a minute ago Up 24 seconds
recipes-user-db-1 postgres:16        "docker-entrypoint.s..." use
r-db             About a minute ago Up About a minute (healthy) 5432/tcp
recipes-user-service-1 recipes-user-service "docker-entrypoint.s..." use
r-service        About a minute ago Up 47 seconds
[root@localhost:/opt/recipes# docker compose -f docker-compose.yml logs --tail 3]
user-service recipe-service social-service bff-service
bff-service-1 | Running server on http://0.0.0.0:3004
recipe-service-1 | query: ALTER TABLE "ingredient" ADD CONSTRAINT "FK_1a884e9b7
0245ac229ded0d8248" FOREIGN KEY ("recipe_id") REFERENCES "recipe"("id") ON DELET
E CASCADE ON UPDATE NO ACTION
recipe-service-1 | query: COMMIT
social-service-1 | RabbitMQ is unavailable, retry in 5 seconds
social-service-1 | RabbitMQ is unavailable, retry in 5 seconds
social-service-1 | Listening to RabbitMQ queue "recipe.deleted"
recipe-service-1 | Data Source has been initialized!
user-service-1 | query: CREATE TABLE "user" ("id" SERIAL NOT NULL, "username"
character varying(50) NOT NULL, "email" character varying(255) NOT NULL, "passw
ord_hash" character varying(255) NOT NULL, "role" "public"."user_role_enum" NOT
```

Рисунок 6. Состояние контейнеров и логи

BFF отвечает 200 на самом сервере. Памяти после запуска хватает благодаря swap.

```
dimaapleev — root@localhost: /opt/recipes — -zsh — 80x36

recipes-user-service-1    recipes-user-service    "docker-entrypoint.s..." use
r-service    About a minute ago    Up 47 seconds
[root@localhost:/opt/recipes# docker compose -f docker-compose.yml logs --tail 3 ]
user-service recipe-service social-service bff-service
bff-service-1 | Running server on http://0.0.0.0:3004
recipe-service-1 | query: ALTER TABLE "ingredient" ADD CONSTRAINT "FK_1a884e9b7
0245ac229ded0d8248" FOREIGN KEY ("recipe_id") REFERENCES "recipe"("id") ON DELET
E CASCADE ON UPDATE NO ACTION
recipe-service-1 | query: COMMIT
social-service-1 | RabbitMQ is unavailable, retry in 5 seconds
social-service-1 | RabbitMQ is unavailable, retry in 5 seconds
social-service-1 | Listening to RabbitMQ queue "recipe.deleted"
recipe-service-1 | Data Source has been initialized!
user-service-1 | query: CREATE TABLE "user" ("id" SERIAL NOT NULL, "username"
character varying(50) NOT NULL, "email" character varying(255) NOT NULL, "passw
ord_hash" character varying(255) NOT NULL, "role" "public"."user_role_enum" NOT
NULL DEFAULT 'USER', "created_at" TIMESTAMP NOT NULL DEFAULT now(), "updated_at"
TIMESTAMP NOT NULL DEFAULT now(), CONSTRAINT "UQ_78a916df40e02a9deb1c4b75edb" U
NIQUE ("username"), CONSTRAINT "UQ_e12875dfb3b1d92d7d7c5377e22" UNIQUE ("email")
, CONSTRAINT "PK_cace4a159ff9f2512dd42373760" PRIMARY KEY ("id"))
user-service-1 | query: COMMIT
user-service-1 | Data Source has been initialized!
[root@localhost:/opt/recipes# curl -s -o /dev/null -w '%{http_code}\n' http://127
.0.0.1:3004/api/v1/recipes
200
[root@localhost:/opt/recipes# free -h
]
              total        used        free      shared  buff/cache   available
Mem:           888Mi        480Mi         73Mi         80Mi        335Mi        181Mi
Swap:          2.0Gi        152Mi         1.9Gi
root@localhost:/opt/recipes# Connection to 140.150.226.124 closed by remote host
.
Connection to 140.150.226.124 closed.
[dimaapleev@MacBook-Air-Dima-2 ~ % ssh root@140.150.226.124
]
[root@140.150.226.124's password:
]
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.0-191-generic x86_64)
```

Рисунок 7. Проверка BFF и память сервера

7. Настройка nginx

Сделана копия конфига. Адрес в `proxy_pass` заменён на `http://127.0.0.1:3004`. Затем конфиг проверен командой `nginx -t` и применён без остановки командой `nginx -s reload`.

```
dimaapleev — root@localhost: /opt/recipes — zsh — 80x36

[dimaapleev@MacBook-Air-Dima-2 ~ % ssh root@140.150.226.124 ]
[root@140.150.226.124's password: ]
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.0-191-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.
New release '24.04.5 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Sat Sep 26 00:13:39 2026 from 185.189.46.58
[root@localhost:~# cd /opt/recipes ]
[root@localhost:/opt/recipes# cp /data/nginx/user_conf.d/django37328.hostkey.in.c]
onf /root/django37328.hostkey.in.conf.bak
[root@localhost:/opt/recipes# sed -i 's#proxy_pass http://django37328.hostkey.in:]
8000;#proxy_pass http://127.0.0.1:3004;#' /data/nginx/user_conf.d/django37328.ho
stkey.in.conf
[root@localhost:/opt/recipes# grep proxy_pass /data/nginx/user_conf.d/django37328]
.hostkey.in.conf
    proxy_pass http://127.0.0.1:3004;
[root@localhost:/opt/recipes# docker exec nginx-nginx-1 nginx -t ]
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
[root@localhost:/opt/recipes# docker exec nginx-nginx-1 nginx -s reload ]
2026/09/25 23:26:02 [notice] 363#363: signal process started
[root@localhost:/opt/recipes# nslookup django37328.hostkey.in ]
-bash: nslookup: command not found
root@localhost:/opt/recipes# Read from remote host 140.150.226.124: Connection r
eset by peer
Connection to 140.150.226.124 closed.
client_loop: send disconnect: Broken pipe
```

Рисунок 8. Изменение и применение конфига nginx

8. Проверка

Запрос к /docs через nginx по HTTPS возвращает 200. Снаружи приложение открывается по IP сервера 140.150.226.124, работают Swagger и запросы к API. Домен из конфига сейчас указывает на другой адрес. На самом сервере домен указывает на 127.0.0.1 (файл /etc/hosts), поэтому проверка с сервера идёт через nginx.

```
[root@localhost:~# cd /opt/recipes ]
[root@localhost:/opt/recipes# curl -sk -o /dev/null -w '%{http_code}\n' --resolve]
django37328.hostkey.in:443:127.0.0.1 https://django37328.hostkey.in/docs/
200
[root@localhost:/opt/recipes# curl -s ifconfig.me; echo ]
140.150.226.124
[root@localhost:/opt/recipes# getent hosts django37328.hostkey.in ]
127.0.0.1      django37328.hostkey.in
[root@localhost:/opt/recipes# grep hostkey /etc/hosts ]
127.0.0.1 django37328.hostkey.in
```

Рисунок 9. Проверка работы через nginx

Вывод

Приложение развёрнуто на удалённом сервере в Docker. Доступ снаружи идёт только через nginx, который передаёт запросы в bff-service.